

Devi Madhuri Donthra Boina

Data Engineer

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SUMMARY

Data Engineer with 5+ years designing, building, and optimizing scalable ETL and ELT pipelines, cloud data platforms, and real time streaming systems across regulated insurance and financial services environments. Strong in Python, PySpark, Scala, and SQL with deep Spark performance tuning, event driven architectures, and data lake and data warehouse design on Snowflake, Redshift, Databricks, and Delta Lake. Builds reliable, observable pipelines with Airflow orchestration, dbt transformations, Terraform infrastructure as code, and data quality and governance built in. Also builds AI and ML data foundations including feature pipelines, RAG, vector databases, and AWS Bedrock. AWS Certified Developer and Cloud Practitioner.

EXPERIENCE

Data Engineer · State Farm, Chicago IL Jul 2024 – Present

- Architected event driven pipelines on AWS Lambda, Step Functions, Kinesis, and EventBridge to process 2+ TB of insurance data daily, with retry logic, dead letter queue routing, and end to end traceability.
- Designed and optimized production ETL pipelines using AWS Glue, EMR, PySpark, Scala, and Python, ingesting from S3, Apache Iceberg, and Glue catalog tables, reducing pipeline runtime 30% through improved partitioning and parallel execution.
- Built the data foundations for ML and GenAI workloads, including feature pipelines and retrieval ready datasets with embeddings and vector stores using AWS Bedrock, RAG, and LangChain.
- Managed cloud infrastructure with Terraform, provisioning queues, DynamoDB, and CloudWatch alerts across environments for reproducible, version controlled deployments.
- Implemented data quality monitoring and anomaly detection with CloudWatch and Python, reducing pipeline errors 25% across 10+ daily runs, and built observability with Splunk that cut incident response time 35%.
- Enforced least privilege IAM security across Lambda, Glue, API Gateway, and cross account S3, configuring STS role assumption in CI/CD to eliminate long lived credentials.

Data Engineer · Bank of America, Hartford CT Aug 2022 – Jun 2024

- Led the end to end migration of the data platform to Delta Lake on Databricks, moving large scale datasets from HDFS and Redshift using AWS Glue and Python automation while preserving full data integrity and ACID compliance, improving processing efficiency 40%.
- Built and maintained production ETL pipelines on AWS Glue, Lambda, and Step Functions to ingest, transform, and load financial data from multiple source systems, supporting 15+ daily scheduled runs.
- Optimized Spark and SQL jobs with partition pruning, broadcast joins, predicate pushdown, and caching, reducing average job runtime 35%, and improved MS SQL Server stored procedures and indexes for faster reporting.
- Designed Snowflake and Redshift data models with dimensional modeling, dynamic masking, and RBAC, lifting query performance 25% while keeping sensitive financial data secured.
- Authored dbt models to centralize transformation logic and implemented Python validation with CloudWatch monitoring to maintain data accuracy across all pipeline runs.

Python Developer · Dentsu, Hyderabad India Aug 2020 – Dec 2021

- Built production ETL pipelines in Python with Pandas, NumPy, and PySpark to ingest and process data from 20+ sources, integrating with Azure Data Factory, Synapse, and ADLS, and GCP BigQuery, Pub/Sub, and Dataflow for scheduling and orchestration.
- Designed and managed multi step workflow orchestration with Apache Airflow using Python based DAGs, reducing manual intervention by automating dependency management across environments.
- Built REST APIs with Flask and Django enabling self service data access for business teams, significantly reducing ad hoc engineering requests, and automated CI/CD with Jenkins and Docker.

EDUCATION

New England College, Henniker NH Jan 2022 – May 2023

M.S. in Computer and Information Systems GPA: 3.95

CVR College of Engineering, India Aug 2016 – Apr 2020

B.Tech in Electronics and Communication Engineering GPA: 3.56

CERTIFICATIONS

- **AWS Certified Developer, Associate** • Issued 2025
- **AWS Certified Cloud Practitioner** • Issued 2023

SKILLS

Python, PySpark, Scala, SQL, Java, Bash, Apache Spark, Spark SQL, Spark Structured Streaming, Kafka, Hadoop, Hive, Apache Iceberg, Airflow, dbt, AWS Glue, EMR, S3, Lambda, Step Functions, Kinesis, Redshift, Athena, CloudWatch, Bedrock, SageMaker, IAM, Azure Data Factory, Synapse, ADLS, Databricks, GCP BigQuery, Pub/Sub, Dataflow, Cloud Composer, Snowflake, Delta Lake, Dimensional Modeling, Star Schema, PostgreSQL, MS SQL Server, MySQL, DynamoDB, MongoDB, Terraform, Docker, Kubernetes, CI/CD, Jenkins, GitLab, GitHub, RBAC, Data Quality, Data Validation, Anomaly Detection, Data Lineage, Observability, Splunk, SLOs, LangChain, RAG, Vector Databases, scikit-learn, MLflow, HuggingFace, Generative AI